1 Experiments

1.1 Dataset

Experiments are conducted on the PetFinder.my Pawpularity dataset, which contains 9912 cat and dog images. In addition to images, meta data is also provided. This tabular meta data has 12 attributes (Subject Focus, Eyes, Face, Near, Action, Accessory, Group, Collage, Human, Occlusion, Info, Blur) and an output Pawpularity score in the range 1–100 for each image. The values of all features are 0 or 1 showing the presence or absence of attribute description. For example, if blur is 1, then it means the pet picture is blur whereas 0 shows the picture is not blur.

1.2 Implementation Detail

All the experiments have been conducted on single NVIDIA GeForce RTX 4080 GPU 16GB with RAM 64 GB. EfficientNet is used as the convolutional baseline, and Swin Transformer applied on image data as the transformer-based backbone. All models are initialized with the pretrained ImageNet weights for vision backbones. Training is performed with Adam optimizers and learning rate scheduling tuned per backbone. Tree-based models for metadata including Gradient Boosting Decision Trees and the Support Vector Regressor (SVR) meta-learner are implemented. Experiments are decomposed to different stages to see the effect of each stage. These stages include images augmentation, backbone change, metadata inclusion via early and late Fusion, models ensembling and stacking. All experiments are conducted using fivefold cross validation. Model performance is evaluated using the root means squared error (RMSE) on out of fold predictions. While for cross validation and meta learner, data is split into train and test and RMSE is calculated on test data. The reason for splitting here is due to each fold model is trained on different fold data, so for ensemble of five 5 folds, we need same to be processed by all fivefold models.

1.3 Pipeline Architecture

The overall architecture as shown in **Figure 1** takes both image data and metadata as inputs. Image data is processed through vision backbones EfficientNet-B4 or the Swin Transformer, which are fine-tuned on the Pawpularity score to extract meaningful visual representations. Metadata, on the other hand, is modeled using a Multi-Layer Perceptron (MLP) for early fusion and a Gradient Boosting Decision Tree (GBDT) for late fusion, alongside a final MLP with two hidden layers. In the early fusion strategy, the vision backbones output embeddings rather than prediction scores, while the MLP transforms metadata into a 64-dimensional embedding. These embeddings are concatenated and passed through additional MLP layers for joint training. In the late fusion strategy, the image models and metadata models are trained independently, and their prediction scores are combined using a weighted average. For cross validation ensemble, the predictions are simple average fivefold models while for the SVR meta learner, predictions are directly predicted from both models prediction as input to SVR.

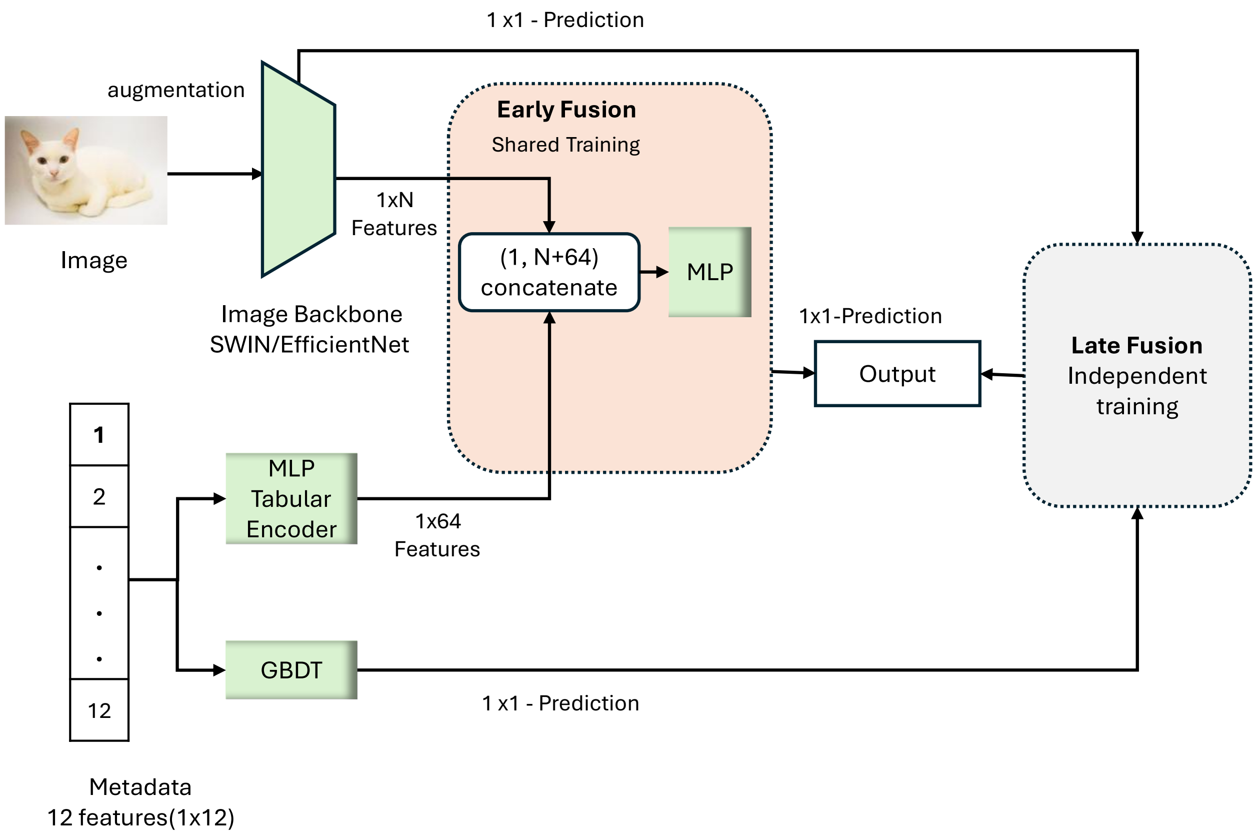

Figure 1 Pipeline

1.4 Augmentation

Dataset augmentation is divided into two categories, light and strong augmentation. For light augmentation, Input Images are randomly flipped horizontally and resized to a fixed resolution. While in the strong augmentation, different variations are added including random Horizontal flipping, rotation, Collor Jittering, Random Affine transformation and random perspective distortions.

1.5 Backbone

We evaluated convolutional and transformer-based backbones. EfficientNetB4 is used as CNN based architecture, while SWIN Transformer is picked as transformer-based model.

1.6 Metadata Fusion

Metadata is incorporated into the architecture through both early and late fusion strategies. In the early fusion approach, tabular metadata is first processed by a Multi-Layer Perceptron (MLP) and projected into a 64-dimensional embedding. This embedding is then concatenated with the feature representations produced by the Swin Transformer or EfficientNet backbone.

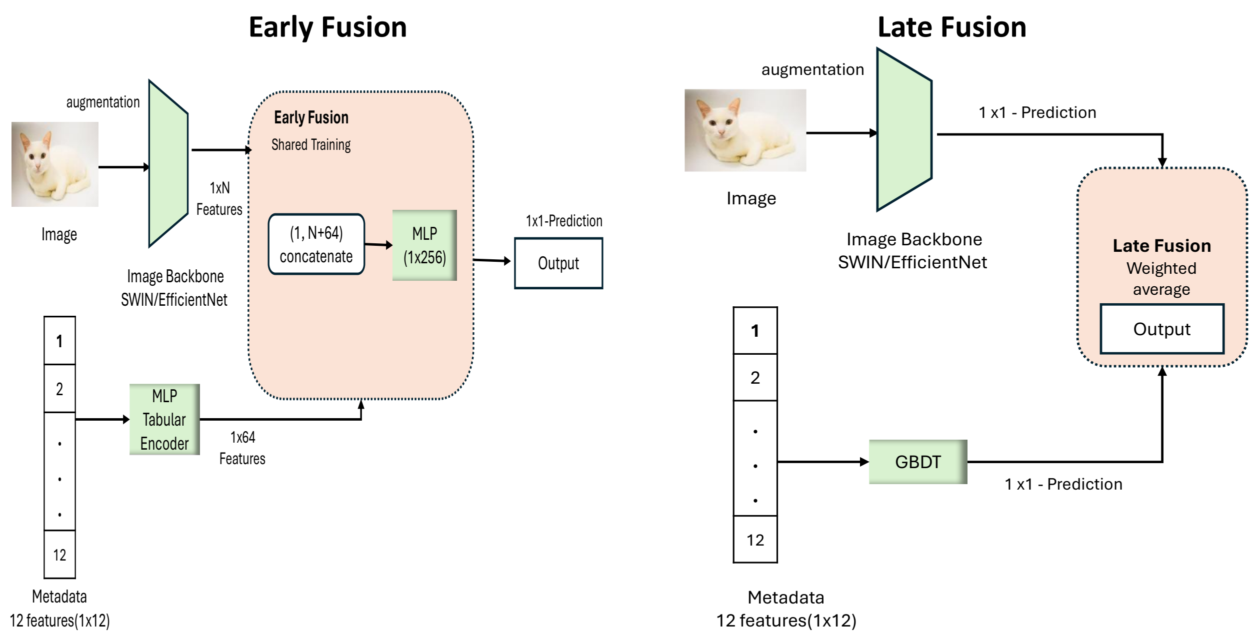

Figure 2 Early and Late Fusion

The fused representation is passed through additional layers, allowing the image models to be trained jointly on both visual features and metadata features. Additionally, the metadata is added into image model by late fusion where both the image model and metadata models are trained independently. Here the metadata is trained with the tree-based models while the image models are trained with SWIN Transformer. At the end, predictions can be merged by weighted average, five folds ensemble or meta learner SVR.

1.7 Image Backbone Late Fusion

We trained the Swin transformer and EfficientNetB4 independently and their predictions are combined via weighted average. Both these models are trained without metadata.

1.8 CV Ensemble

We assessed the single best model out of five folds as well as five models ensemble by average to see the individual as well as collective effect of all folds. Unlike other stages, the data is split to train and test with the ratio of 70% and 30%.

1.9 Meta Model Learner (Stacked model)

In this experiment, a Swin Transformer model is trained on image data, while a Gradient Boosting Decision Tree (GBDT) model is trained on metadata using 70% of the training set. Five-fold cross-validation is applied, and out-of-fold (OOF) predictions are collected from both models. These OOF predictions are then stacked to form the input for a second-stage Support Vector Regressor (SVR) meta-learner, which is trained to produce the final predictions. During inference, the test data is evaluated by all five-fold-specific models of each backbone. The predictions from the five folds are averaged, resulting in two features swin mean and gbdt mean. The final predictions are obtained by feeding these predictions as input into the second-stage SVR model.

2 Results and Ablation Study

This table overviews the stage-wise ablation experiments for the multimodal Pawpularity pipeline. For each stage, the table lists the vision backbone, augmentation setting, use of tabular metadata, fusion type, and ensemble configuration, together with the mean RMSE and standard deviation over five folds. The ΔRMSE column reports the change in RMSE within each stage when moving from the first to the second configuration. Each stage is analyzed in more detail in the following subsections.

Table 1 Stage wise ablations results

Stage	Vision	Augment	Tabular	Fusion	Ensemble	RMSE	Std	ΔRMSE
1a:Aug	EffNet-B4	Light	–	–	–	19.06	0.43	-0.38
	EffNet-B4	Strong	–	–	–	18.68	0.29	
1b:Aug	SWIN	Light	–	–	–	17.71	0.33	-0.14
	SWIN	Strong	–	–	–	17.86	0.30	
2:backbone	EffNet-B4	Strong	–	–	–	18.68	0.29	-0.96
	Swin-T	Strong	–	–	–	17.71	0.33	
3a:EarlyFusion	Swin-T	Strong		–	–	17.71	0.33	-0.11
	Swin-T	Strong	MLP	Early	–	17.60	0.16	
3b:LateFusion	Swin-T	Strong		–	–	17.71	0.33	-0.07
	Swin-T	Strong	GBDT	Late	–	17.64	0.31	
3c:EarlyLateFusion	Swin-T	Strong	MLP	Early	–	17.60	0.16	+0.04
	Swin-T	Strong	GBDT	Late	–	17.64	0.31	
4:Backbone ens.	Swin-T	Strong	–	–	–	17.71	0.33	-0.11
	Sw+EfB4	Strong	–	–		17.59	0.29	
5a:Ensemble	Swin-T	Strong		Early	Bestfold	17.69	0.16	-0.19
	Swin-Ens	Strong		Early	5-fold	17.49	0.16	
5b:Stacking	Swin-Ens	Strong	MLP	Early	5folds	17.49	0.16	0.11
	Swin-Ens	Strong	GBDT	Early	SVR-stack	17.61		

2.1 Augmentation

The efficientB4 model RMSE is decreased by 0.38 when high resolution with strong augmentation is applied collectively. However, SWIN is not much affected by the Resolution and Augmentation. This is because EfficientNet is CNN-based and resolution sensitive while Swin is patch-based and already models local and global context.

2.2 Backbone

The most impactful component is Backbone change from EfficientNet to Swin model where the RMSE dropped by 0.96 compared to EfficientNet. Swin Outperformed the CNN based model, because it can see the image as a set of patches and uses multi-head self-attention to capture long-range dependencies whereas EfficientNet relies on local convolutions and pooling, which are less expressive for scene-level images

2.3 Metadata Fusion

The early metadata infusion and late fusion are analyzed. However, the metadata does not have enough signal and contains binary values with very weak correlations with target, so inclusion of metadata does not improve the overall performance and the RMSE is slightly dropped by 0.07 and 0.11 for early and late fusion respectively.

2.4 Image Backbone Late Fusion

Late fusion of EfficientNetB4 and Swin transformers give the negligible RMSE difference of 0.11. Even though they both have different architectures, however this small gain indicates that both backbones share similar error patterns on Pawpularity prediction.

2.5 CV Ensemble and second stage learner (stacked model)

Similarly, the CV ensemble and second-stage SVR meta-learner do not able to consistent improvements over the best single Swin model. Because all image and metadata models exhibit similar high errors in the lowest and highest Pawpularity ranges

2.6 Error Patterns Analysis

The most influential stage is backbone selection because changing the backbone to swin transformer produces the largest performance shift (0.96) in the entire stage-wise analysis. This suggests that Pawpularity prediction performance depends primarily on the quality of the Swin image only representation, its ability to capture both local and broader scene context rather than metadata and fusion strategies. However, from Figure 3, it can be seen that the vision models are producing high errors in the lower and high pawpularity regions Even though the Swin Transformer achieves lower average RMSE than EfficientNet-B4 overall, the bin-wise analysis shows that both backbones still produce their largest errors in the sparsely populated Pawpularity ranges (0–20 and 60–100). The mid-range scores have many training examples, so both models learn stable representations there, whereas the extreme low and high scores regions have less records, leading to underfitting in the tails.

Table 2 where the number of records is less which makes it difficult to capture enough patterns for high and lower pawpularity animals.

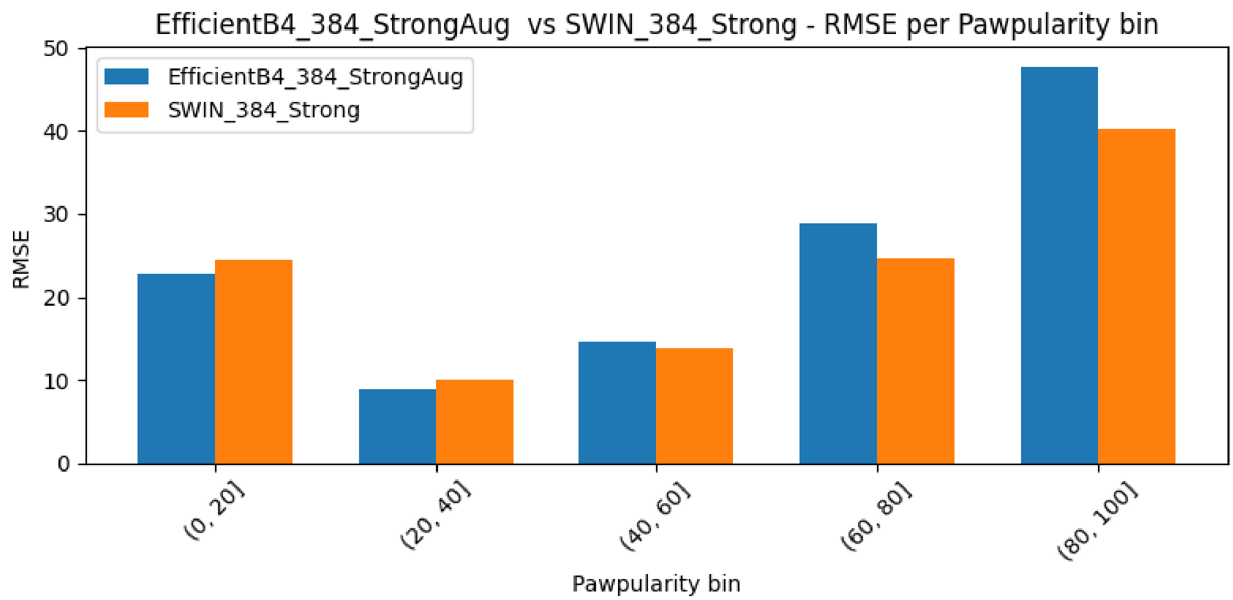

Figure 3 Errors Distribution

Even though the Swin Transformer achieves lower average RMSE than EfficientNet-B4 overall, the bin-wise analysis shows that both backbones still produce their largest errors in the sparsely populated Pawpularity ranges (0–20 and 60–100). The mid-range scores have many training examples, so both models learn stable representations there, whereas the extreme low and high scores regions have less records, leading to underfitting in the tails.

Table 2 Data Distribution and Errors Regions

Bin	Data Count	EffNet-B4 RMSE	Swin-T RMSE	ΔRMSE (Swin − EffNet)
(0, 20]	1417	22.8155	24.4964	+1.6809
(20, 40]	5119	8.8918	9.9702	+1.0785
(40, 60]	2088	14.5251	13.8592	-0.6658
(60, 80]	727	28.7612	24.5777	-4.1835
(80, 100]	561	47.6788	40.3003	-7.3785